Games, Payoffs and Utilities

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What is a Game?

Plan

- We've covered logic, probability, and individual decision-making.
- Now we turn to **game theory**: decisions where outcomes depend on what *other rational agents* choose.
- Today we're looking at what games are, how to represent them, and when one strategy clearly dominates another.

Associated Reading

Game Theory, Sections 2.1 and 2.2

A Motivating Example

A British TV game show, *Golden Balls*, ends with a simple game called **Split or Steal**.

Two players, Sarah and Steven, each secretly pick one of two balls:

- The **Split** ball, or
- The **Steal** ball.

The jackpot is \$100,000. What happens next depends on their choices.

	Steven: Split	Steven: Steal
Sarah: Split	Each gets \$50,000	Sarah: \$0, Steven: \$100,000
Sarah: Steal	Sarah: \$100,000, Steven: \$0	Both get nothing

This is a **game-frame**: we know the players, the strategies, and the outcomes, but not what the players *prefer*.

Game-Frame vs. Game

A **game-frame** specifies:

- The set of **players**
- The **strategies** available to each player
- The **outcomes** that result from each combination of strategies

But to know what rational players will *do*, we also need to know their **preferences** over outcomes.

A **game** is a game-frame plus a ranking of outcomes for each player.

What Should Sarah Do?

Suppose both players are **selfish and greedy**: they only care about their own money, and prefer more to less.

Sarah reasons:

- If Steven picks **Steal**, it doesn't matter what she does. She gets nothing either way.
- If Steven picks **Split**, she gets $50k from Split or $100k from Steal.

So: if greedy, **Steal weakly dominates Split**.

But crucially, this is not the right analysis for *every* player.

What If Sarah Is Fair-Minded?

Suppose instead Sarah prefers both getting \$50k to her getting \$100k alone. She cares about fairness.

Her ranking is: (both \$50k) $\succ$ (she gets \$100k) $\succ$ (Steven gets \$100k) $\succ$ (both get nothing).

Now:

- If Steven Splits: Sarah prefers Split (\$50k each) over Steal (\$100k for her alone).
- If Steven Steals: Sarah prefers Split (at least Steven gets something) over Steal (nothing for anyone).

Split now strictly dominates Steal.

With the same physical outcomes, different preferences lead to **different rational strategies**.

This is why game theory requires us to specify preferences carefully. The numbers in a payoff table aren't money; they're **utilities**, capturing everything a player values.

"Selfish and greedy" is a *model assumption*, not a universal truth.

Many people are strongly motivated by fairness, and practically everyone has declining marginal utility for money.

Terminology

I’ve used two bits of terminology there: **strictly dominates** and **weakly dominates**.

Option A **strictly dominates** Option B if it does better no matter what the other player does (or what the other players do).

Option B **weakly dominates** Option B if might do better and can’t do worse.

The difference is that weak dominance allows for **ties**. We’ll say more about these notions later.

Representing Games

To work with preferences mathematically, we assign numbers to outcomes so that higher numbers mean more preferred.

Definition: An *ordinal utility function* U assigns a number $U(o)$ to each outcome o such that $U(o) > U(o')$ if and only if outcome o is preferred to o'.

When we are working with ordinal utility, the *magnitudes* don't matter, only the *ordering*. We'll come back to cases where the magnitudes do matter; but it's good to start just with rankings.

With selfish/greedy preferences, assign utilities so that getting $100k alone = 4, both get $50k = 3, other outcomes = 2:

	Steven: Split	Steven: Steal
Sarah: Split	(3, 3)	(2, 4)
Sarah: Steal	(4, 2)	(2, 2)

Convention: **(Row player, Column player)** in each cell.

With Fair-Minded Preferences

If Sarah is fair-minded (prefers (both $50k) above all else), her utilities shift:

	Steven: Split	Steven: Steal
Sarah: Split	(4, 3)	(2, 4)
Sarah: Steal	(3, 2)	(1, 2)

Same game-frame, same second player, but different game because Sarah's ranking has changed.

The Normal (Strategic) Form

A two player game in **normal form** (or *strategic form*) is a table where:

- **Rows** = strategies for Player 1
- **Columns** = strategies for Player 2
- **Cells** = payoffs (utilities) for both players

This representation is called *normal form* or *strategic form*. It captures all the strategically relevant information when players move simultaneously.

The textbook's central example uses a different story. (This is on page 19.)

Two co-workers, Doug and Ed, are competing for a prize. Each can put in Normal or Extra effort.

	Ed: Normal	Ed: Extra
Doug: Normal	(2, 2)	(0, 3)
Doug: Extra	(3, 0)	(1, 1)

Each would rather get the prize (at cost of family time) than have the other get it. Extra effort is the only way to secure it.

The Prisoner's Dilemma Tension

	Ed: Normal	Ed: Extra
Doug: Normal	(2, 2)	(0, 3)
Doug: Extra	(3, 0)	(1, 1)

Regardless of what Ed does, Doug does better with Extra effort: $3 > 2$ and $1 > 0$. And the same reasoning applies to Ed.

But if both choose Extra, they both get 1. This is worse than both choosing Normal (both get 2).

Dominance

Strict Dominance

Strategy a **strictly dominates** strategy b for Player i if a gives Player i a strictly *higher* payoff than b in *every possible situation*, no matter what the other players do.

Formal statement For every strategy profile of the opponents,

$$\pi_i(a, s_{-i}) > \pi_i(b, s_{-i}).$$

It seems plausible to say that a rational player will never play a strictly dominated strategy, though we'll have to be careful to think about what game they're playing if we want to check whether that's always true.

Strict Dominance in the PD

	Ed: Normal	Ed: Extra
Doug: Normal	(2, 2)	(0, 3)
Doug: Extra	(3, 0)	(1, 1)

Does Extra strictly dominate Normal for Doug?

- Against Ed: Normal → Doug gets 3 (Extra) vs 2 (Normal)
- Against Ed: Extra → Doug gets 1 (Extra) vs 0 (Normal)

Yes; **Extra strictly dominates Normal** for Doug. By symmetry, same for Ed.

Strategy a **weakly dominates** strategy b if:

- a gives Player i a payoff **at least as good** as b in every situation, **and**
- a gives a **strictly better** payoff in at least one situation.

Weak dominance is a *weaker* condition. It rules out some strategies, but is somewhat more controversial.

In the Split/Steal game with selfish preferences: **Steal weakly dominates Split** (same payoff when opponent steals; better when opponent splits).

Strategy a is a **strictly dominant strategy** if it strictly dominates every other strategy.

Strategy a is a **weakly dominant strategy** if it weakly dominates every other strategy (and isn't equivalent to it).

In the PD, Extra effort is a *strictly* dominant strategy for each player.

In the selfish Split/Steal game, Steal is only a *weakly* dominant strategy.

A strategy profile is a **strict dominant-strategy equilibrium** if every player is using a strictly dominant strategy.

A strategy profile is a **weak dominant-strategy equilibrium** if every player is using a weakly dominant strategy, and at least one player's strategy is only weakly (not strictly) dominant.

In the PD, (Extra, Extra) is a **strict dominant-strategy equilibrium**.

Pareto Superiority

Outcome o is **strictly Pareto superior** to o' if *every* player prefers o to o'.

In the PD: (Normal, Normal) is strictly Pareto superior to (Extra, Extra).

This is the coordination failure in the PD:

- Individual rationality leads each player to choose Extra effort.
- The resulting outcome (1, 1) is strictly Pareto inferior to (2, 2).
- Rational individual choice produces a collectively bad outcome.

The PD structure models any situation where:

- There is a cooperative outcome that is good for everyone.
- But each individual is better off defecting, regardless of what others do.

Possible examples: arms races, price competition, environmental agreements, overexploiting shared resources, doping in sport.

Example

Consider the following two-player game (payoffs shown as (Player 1, Player 2)):

	L	C	R
U	(3, 2)	(1, 1)	(2, 3)
M	(1, 0)	(2, 2)	(1, 1)
D	(3, 1)	(1, 3)	(3, 2)

What are the dominance relations between the strategies?

Row

	L	C	R
U	(3, 2)	(1, 1)	(2, 3)
M	(1, 0)	(2, 2)	(1, 1)
D	(3, 1)	(1, 3)	(3, 2)

Does **U** dominate **M**?

- If Column plays **L**, then **U** gets 3, **M** gets 1.
- If Column plays **C**, then **U** gets 1, **M** gets 2.

We don't need to check the last column; there is no dominance relation here.

Row

	L	C	R
U	(3, 2)	(1, 1)	(2, 3)
M	(1, 0)	(2, 2)	(1, 1)
D	(3, 1)	(1, 3)	(3, 2)

Does M dominate U?

- If Column plays L, then M gets 1, U gets 3. ? **No dominance**

	L	C	R
U	(3, 2)	(1, 1)	(2, 3)
M	(1, 0)	(2, 2)	(1, 1)
D	(3, 1)	(1, 3)	(3, 2)

Does **U** dominate **D**?

- If Column plays **L**, then **U** gets 3, **D** gets 3. =
- If Column plays **C**, then **U** gets 1, **D** gets 1. =
- If Column plays **R**, then **U** gets 2, **D** gets 3. $\boxed{?}$

No dominance

	L	C	R
U	(3, 2)	(1, 1)	(2, 3)
M	(1, 0)	(2, 2)	(1, 1)
D	(3, 1)	(1, 3)	(3, 2)

Does **D** dominate **U**?

- If Column plays **L**, then **D** gets 3, **D** gets 3. =
- If Column plays **C**, then **D** gets 1, **D** gets 1. =
- If Column plays **R**, then **D** gets 3, **D** gets 2. ?

It's **never worse** and **sometimes better** so **weak dominance**.

Row

	L	C	R
U	(3, 2)	(1, 1)	(2, 3)
M	(1, 0)	(2, 2)	(1, 1)
D	(3, 1)	(1, 3)	(3, 2)

Does **M** dominate **D**?

- If Column plays **L**, then **M** gets 1, **D** gets 3. ?

No dominance

	L	C	R
U	(3, 2)	(1, 1)	(2, 3)
M	(1, 0)	(2, 2)	(1, 1)
D	(3, 1)	(1, 3)	(3, 2)

Does **D** dominate **M**?

- If Column plays **L**, then **D** gets 3, **M** gets 1.
- If Column plays **C**, then **D** gets 1, **M** gets 2.

No dominance

Column

	L	C	R
U	(3, 2)	(1, 1)	(2, 3)
M	(1, 0)	(2, 2)	(1, 1)
D	(3, 1)	(1, 3)	(3, 2)

Does L dominate C?

- If Row plays U: L gets 2, C gets 1.
- If Row plays M: L gets 0, C gets 2.

No dominance

	L	C	R
U	(3, 2)	(1, 1)	(2, 3)
M	(1, 0)	(2, 2)	(1, 1)
D	(3, 1)	(1, 3)	(3, 2)

Does **C** dominate **L**?

- If Row plays **U**: C gets 1, L gets 2. ? **No dominance**

Column

	L	C	R
U	(3, 2)	(1, 1)	(2, 3)
M	(1, 0)	(2, 2)	(1, 1)
D	(3, 1)	(1, 3)	(3, 2)

Does **L** dominate **R**?

- If Row plays **U**: L gets 2, R gets 3. **No dominance**

Column

	L	C	R
U	(3, 2)	(1, 1)	(2, 3)
M	(1, 0)	(2, 2)	(1, 1)
D	(3, 1)	(1, 3)	(3, 2)

Does **R** dominate **L**?

- If Row plays **U**: R gets 3, L gets 2.
- If Row plays **M**: R gets 1, L gets 0.
- If Row plays **D**: R gets 2, L gets 1.

R strictly dominates L

	L	C	R
U	(3, 2)	(1, 1)	(2, 3)
M	(1, 0)	(2, 2)	(1, 1)
D	(3, 1)	(1, 3)	(3, 2)

Does C dominate R?

- If Row plays U: C gets 1, R gets 3. ? **No dominance**

Column

	L	C	R
U	(3, 2)	(1, 1)	(2, 3)
M	(1, 0)	(2, 2)	(1, 1)
D	(3, 1)	(1, 3)	(3, 2)

Does **R** dominate **C**?

- If Row plays **U**: R gets 3, C gets 1.
- If Row plays **M**: R gets 1, C gets 2.

No dominance

- For Row: **D weakly dominates U**. No other dominance relations.
- For Column: **R strictly dominates L**. No other dominance relations.

For Next Time

- Dominant strategies don't always exist.
- To solve more games, we'll put two more concepts on the table:

1. **Nash equilibrium**; and
2. **iterated elimination**.